Design of Lead-Lag Compensation via Root-Locus
— Lead Compensation

(通过根轨迹的超前-滞后补偿设计 — 超前补偿器)

Lead-Lag Compensation

- **General Form (一般形式)**

$$\frac{s+\beta}{s+\alpha}$$

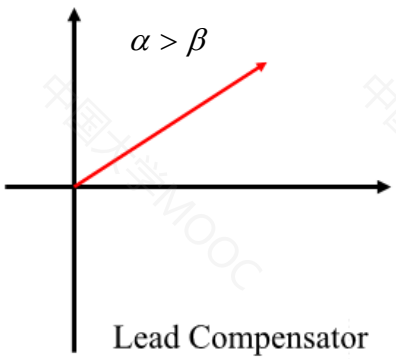
If β<α, it is a lead compensator (high-pass)

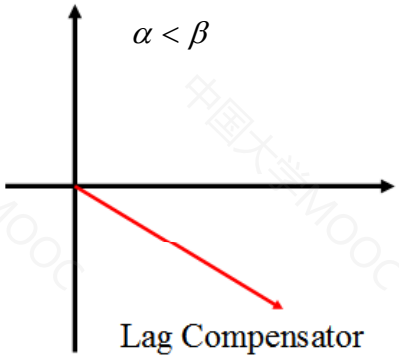
If β>α, it is a lag compensator (low-pass)

- **Intuitive Interrelation (直观解释)**
  - Frequency Response

$$\left.\frac{s+\beta}{s+\alpha}\right|_{s=j\omega}=\frac{j\omega+\beta}{j\omega+\alpha}=\frac{\alpha\beta+\omega^2+j(\alpha-\beta)\omega}{\omega^2+\alpha^2}$$

$\Rightarrow$ Phase Angle:



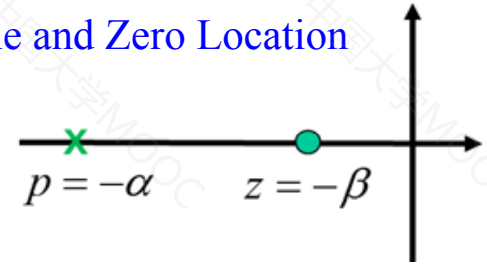


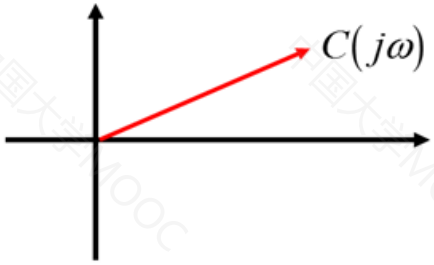
Lead-Lag Compensation

Lead Compensator (超前控制器)

$$C(s) = K_c \frac{s+\beta}{s+\alpha}, \quad \beta < \alpha$$

Pole and Zero Location



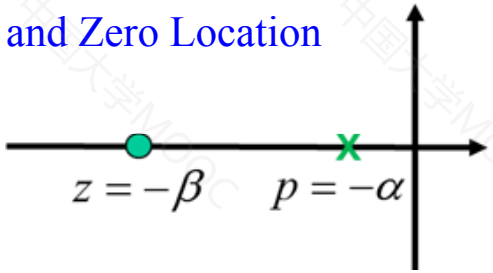


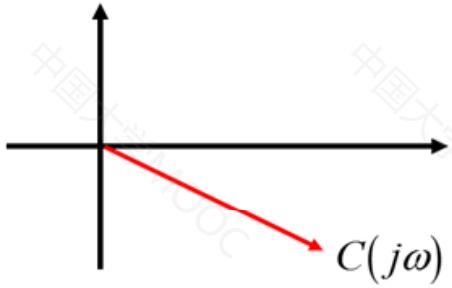
Phase Lead

Lag Compensator (滞后控制器)

$$C(s) = K_c \frac{s+\beta}{s+\alpha}, \quad \beta > \alpha$$

Pole and Zero Location



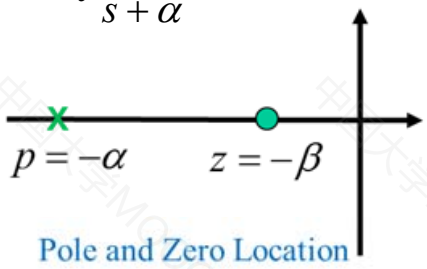


Phase Lag

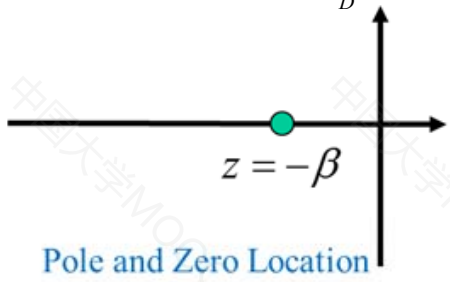
Lead-Lag Compensation and PD/PI Control

Lead Controller (超前控制器) $\xrightarrow{\alpha \to \infty}$ **PD Controller (PD控制器)**

$$C(s) = K_c \frac{s+\beta}{s+\alpha}, \quad \beta < \alpha$$

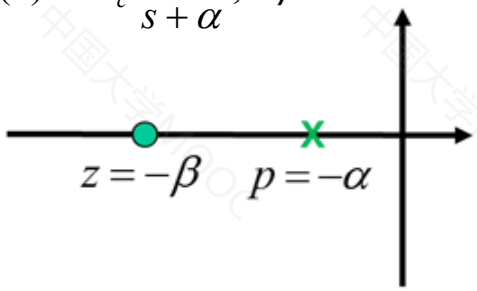


$$C(s) = K_P + K_D s = K_D(s+\beta), \quad \beta = \frac{K_P}{K_D}$$

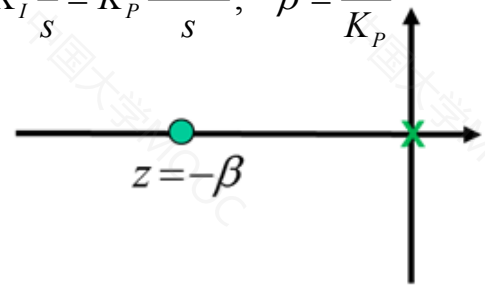


Lag Controller (滞后控制器) $\xrightarrow{\alpha \to 0}$ **PI Controller (PI控制器)**

$$C(s) = K_c \frac{s+\beta}{s+\alpha}, \quad \beta > \alpha$$



$$C(s) = K_P + K_I \frac{1}{s} = K_P \frac{s+\beta}{s}, \quad \beta = \frac{K_I}{K_P}$$

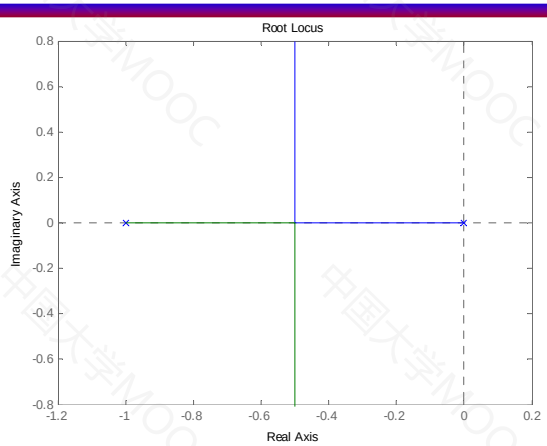


Lead Compensation

Ex. Add lead compensator to increase system damping (增加超前补偿器来增加系统阻尼)

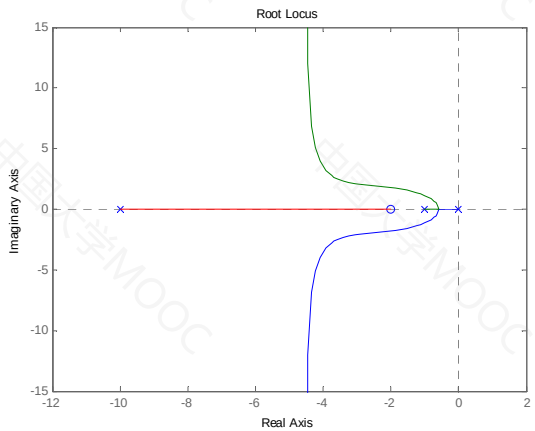
$$\frac{N(s)}{D(s)} = \frac{1}{s(s+1)}$$

```
>> op_num=[1];
>> op_den=[1,1,0];
>> rlocus(op_num,op_den);
```



$$\frac{N(s)}{D(s)} = \frac{1}{s(s+1)}\frac{s+2}{s+10}$$

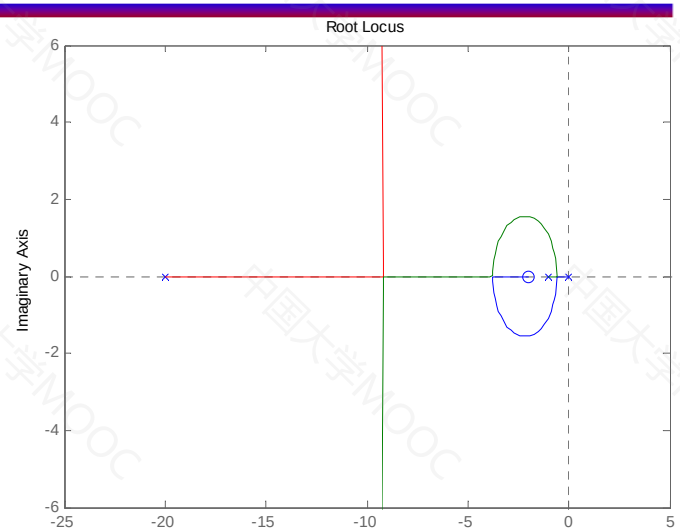
```
>> op_num=[1,2];
>> op_den=[1,11,10,0];
>> rlocus(op_num,op_den);
```



Lead Compensation

$$\frac{N(s)}{D(s)} = \frac{1}{s(s+1)}\frac{s+2}{s+20}$$

```
>> op_num=[1,2];
>> op_den=[1,21,20,0];
>> rlocus(op_num,op_den);
```



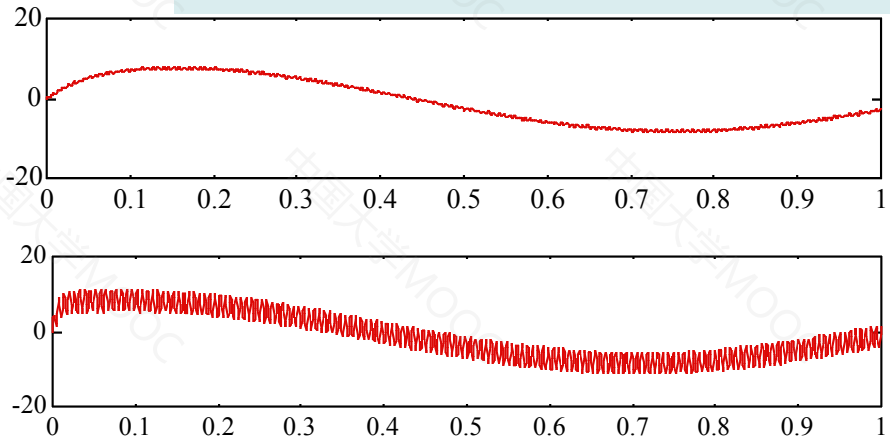
- The added zero at -2 pulls the root locus to the left (在-2处增加的零点将根轨迹拉向左边)
  - ⇒ The closed-loop system more stable (闭环系统更稳定)
- The further left we place the added zero (在更左边配置增加的零点)
  - ⇒ The more attraction the added zero will have (增加的零点将更具吸引性)
- If the added pole to -∞ (如果增加的极点在-∞)
  - ⇒ Ideal PD controller (理想PD控制器)
- It should not be overdue to place the pole farther to the left (不要将极点配置得太靠左)

Lead Compensation

Ex. The following plots are obtained using MATLAB for two compensators $31\frac{s+2}{s+20}$ and $310\frac{s+2}{s+200}$ assuming a 1% high frequency noise in $e(t)$

```
num=[1 2]*31; den=[1 20];;  %Assume K=31
t=0:0.001:1; e=sin(5*t)+0.01*sin(1000*t);
[u,x]=lsim(num,den,e,t);
plot(t,u); pause
num=[1 2]*310; den=[1 200];
[u,x]=lsim(num,den,e,t);
plot(t,u); axis
```

The same static gain and zero for both compensator, a much severe control input chattering for the second second compensator due to faster pole (at –200) (两个补偿器具有相同的静态增益和零点，但第二个补偿器因有较快的极点导致更严重的控制量振荡)



Zero and Pole Selection of Lead Compensation

Trial and error process of selecting exact values of β and α :

- the zero β is placed in the neighborhood of the closed-loop ω_n, as determined by rise-time or settling time requirements (零点β应放置在闭环ω_n的附近,以满足上升时间或调整时间的要求)
- the poles α is located at a distance 3 to 20 times the value of the zero location (极点α应放置在离零点3-20倍的距离).

A compromise between the conflicting effects of noise suppression and compensation effectiveness!
(在噪声抑制和补偿效果的冲突之间获得妥协)

Notes :

- if the pole is too close to the zero, the root locus moves back too far toward its uncompensated shape and the zero is not successful in doing its job of bending the root locus toward its location to add (如果极点太靠近零点，根轨迹会过多移回到未补偿的形状，而零点没有足够能力将根轨迹向增加的位置移动)
- If the pole is too far to the left, the magnification of noise at the output of the controller would be too great, which may lead the actuator of the process overheated by noise energy (如果极点左移太多，控制器输出的噪声将会太大).

Procedure of Designing Lead Compensation

General guideline of designing lead compensation (设计超前补偿器的一般原则):

- Choose the desired closed-loop pole location for the dominant poles (the pole (poles) which is closest to the imaginary axis) (选择期望闭环极点的主导极点(离虚轴最近的极点)).
- Arbitrarily pick one lead-compensation parameter and use the Phase Condition to select the other (任意选择超前控制器参数，并应用相角条件选择其它参数).
  - ✓ *pick β location on the real axis straight below the desired pole location or to the left* (选择β直接放置在期望极点或以左的实轴上)
  - ✓ *calculate the location of α using the Phase Condition. Plot the final root locus* (运用相角条件计算α位置，画出最终根轨迹)
  - ✓ *calculate the gain K using Magnitude Condition* (运用幅值条件计算增益*K*).
- Several iterations of this procedure before the final design (在最终确定前进行多次迭代尝试)
  - ✓ *p=-α too far to the left, other branches becoming dominant* (*p=-α*太靠左，其他分支成为主导),
  - ✓ *steady-state error too large, etc* (稳态误差太大).
- Relax the requirements on steady-state error since the steady-state error can be reduced by a lag compensation (因滞后补偿器会降低稳态误差，可对稳态误差放宽要求).

Summary

- **Structure of Lead-Lag Compensation** (超前-滞后补偿器结构)
- **Design of Lead Compensation** (超前补偿器设计)
  - ✓ Zero and Pole Selection of Lead Compensation (超前补偿器的零极点选择)
  - ✓ Procedure of Designing Lead Compensation (设计超前补偿器的步骤)